

VGG-16 and VGG-19 Architecture

What is VGG?

VGG refers to a series of deep convolutional neural networks developed by the Visual Geometry Group (VGG) at the University of Oxford, introduced in the 2014 paper "Very Deep Convolutional Networks for Large-Scale Image Recognition" by Simonyan and Zisserman. VGG models gained popularity for their simplicity and high performance in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2014.

Key Features of VGG Networks

- **Uniform Architecture:** All convolution layers use 3×3 kernels with stride 1 and padding 'same'.
 - **Max Pooling:** 2×2 windows with stride 2 follow some conv layers to reduce spatial dimensions.
 - **Depth:** VGG-16 has 16 layers with learnable parameters (13 conv + 3 FC), while VGG-19 has 19 (16 conv + 3 FC).
 - **High Performance:** Despite its simplicity, VGG achieved top-tier accuracy in classification tasks.
 - **Input Image Size:** $224 \times 224 \times 3$ (RGB image).
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VGG-16 Architecture (Detailed Layer-wise Explanation)

VGG-16 Architecture Breakdown

1. Input Layer:

- Input Image Size: $(224 \times 224 \times 3)$ (RGB image)

2. Convolutional Layers:

Conv1 Block:

- Conv1-1: 64 filters of size 3×3 , stride 1, padding = 'same' → Output: $224 \times 224 \times 64$
- Conv1-2: 64 filters of size 3×3 , stride 1, padding = 'same' → Output: $224 \times 224 \times 64$

- Max Pooling (Pool1): 2x2 filter, stride 2 → Output: 112 x 112 x 64

Conv2 Block:

- Conv2-1: 128 filters of size 3x3, stride 1, padding = 'same' → Output: 112 x 112 x 128
- Conv2-2: 128 filters of size 3x3, stride 1, padding = 'same' → Output: 112 x 112 x 128
- Max Pooling (Pool2): 2x2 filter, stride 2 → Output: 56 x 56 x 128

Conv3 Block:

- Conv3-1: 256 filters of size 3x3, stride 1, padding = 'same' → Output: 56 x 56 x 256
- Conv3-2: 256 filters of size 3x3, stride 1, padding = 'same' → Output: 56 x 56 x 256
- Conv3-3: 256 filters of size 3x3, stride 1, padding = 'same' → Output: 56 x 56 x 256
- Max Pooling (Pool3): 2x2 filter, stride 2 → Output: 28 x 28 x 256

Conv4 Block:

- Conv4-1: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 28 x 28 x 512
- Conv4-2: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 28 x 28 x 512
- Conv4-3: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 28 x 28 x 512
- Max Pooling (Pool4): 2x2 filter, stride 2 → Output: 14 x 14 x 512

Conv5 Block:

- Conv5-1: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 14 x 14 x 512
- Conv5-2: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 14 x 14 x 512
- Conv5-3: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 14 x 14 x 512
- Max Pooling (Pool5): 2x2 filter, stride 2 → Output: 7 x 7 x 512

3. Fully Connected Layers:

- Flatten: Converts 3D to 1D → $7 \times 7 \times 512 = 25088$
- FC6: Fully connected layer with 4096 neurons → Output: 4096
- FC7: Fully connected layer with 4096 neurons → Output: 4096
- FC8: Fully connected layer with 1000 neurons (for 1000 ImageNet classes) → Output: 1000

4. Output Layer:

- Softmax activation to produce probability distribution over 1000 classes

VGG-19 Architecture Breakdown

VGG-19 is an extension of VGG-16, adding more convolutional layers in Conv3, Conv4, and Conv5 blocks.

1. Input Layer:

- Input Image Size: (224 x 224 x 3)

2. Convolutional Layers:

Conv1 Block:

- Conv1-1: 64 filters of size 3x3, stride 1, padding = 'same' → Output: 224 x 224 x 64
- Conv1-2: 64 filters of size 3x3, stride 1, padding = 'same' → Output: 224 x 224 x 64
- Max Pooling (Pool1): 2x2 filter, stride 2 → Output: 112 x 112 x 64

Conv2 Block:

- Conv2-1: 128 filters of size 3x3, stride 1, padding = 'same' → Output: 112 x 112 x 128
- Conv2-2: 128 filters of size 3x3, stride 1, padding = 'same' → Output: 112 x 112 x 128
- Max Pooling (Pool2): 2x2 filter, stride 2 → Output: 56 x 56 x 128

Conv3 Block:

- Conv3-1: 256 filters of size 3x3, stride 1, padding = 'same' → Output: 56 x 56 x 256
- Conv3-2: 256 filters of size 3x3, stride 1, padding = 'same' → Output: 56 x 56 x 256
- Conv3-3: 256 filters of size 3x3, stride 1, padding = 'same' → Output: 56 x 56 x 256
- Conv3-4: 256 filters of size 3x3, stride 1, padding = 'same' → Output: 56 x 56 x 256
- Max Pooling (Pool3): 2x2 filter, stride 2 → Output: 28 x 28 x 256

Conv4 Block:

- Conv4-1: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 28 x 28 x 512
- Conv4-2: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 28 x 28 x 512
- Conv4-3: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 28 x 28 x 512
- Conv4-4: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 28 x 28 x 512
- Max Pooling (Pool4): 2x2 filter, stride 2 → Output: 14 x 14 x 512

Conv5 Block:

- Conv5-1: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 14 x 14 x 512
- Conv5-2: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 14 x 14 x 512
- Conv5-3: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 14 x 14 x 512
- Conv5-4: 512 filters of size 3x3, stride 1, padding = 'same' → Output: 14 x 14 x 512

- Max Pooling (Pool5): 2x2 filter, stride 2 → Output: 7 x 7 x 512

3. Fully Connected Layers:

- Flatten: Converts 3D to 1D → $7 \times 7 \times 512 = 25088$
- FC6: Fully connected layer with 4096 neurons → Output: 4096
- FC7: Fully connected layer with 4096 neurons → Output: 4096
- FC8: Fully connected layer with 1000 neurons (for 1000 ImageNet classes) → Output: 1000

4. Output Layer:

- Softmax activation to produce probability distribution over 1000 classes

Tabular Summary of VGG-16

Layer Type	Output Size	Kernel Size	Filters	Stride	Activation
Input	224×224×3	-	-	-	-
Conv1_1	224×224×64	3×3	64	1	ReLU
Conv1_2	224×224×64	3×3	64	1	ReLU
MaxPool1	112×112×64	2×2	-	2	-
Conv2_1	112×112×128	3×3	128	1	ReLU
Conv2_2	112×112×128	3×3	128	1	ReLU
MaxPool2	56×56×128	2×2	-	2	-
Conv3_1/2/3	56×56×256	3×3	256	1	ReLU
MaxPool3	28×28×256	2×2	-	2	-
Conv4_1/2/3	28×28×512	3×3	512	1	ReLU
MaxPool4	14×14×512	2×2	-	2	-
Conv5_1/2/3	14×14×512	3×3	512	1	ReLU
MaxPool5	7×7×512	2×2	-	2	-
Flatten	25088	-	-	-	-
FC1	4096	-	-	-	ReLU
FC2	4096	-	-	-	ReLU
FC3	1000	-	-	-	Softmax

Development History

- **2014:** VGGNet introduced at ILSVRC.
 - **VGG-16 & VGG-19:** Gained prominence due to simplicity and transfer learning capability.
 - **Deeper Models:** Inspired deeper CNNs like ResNet by showing depth improves performance.
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Applications

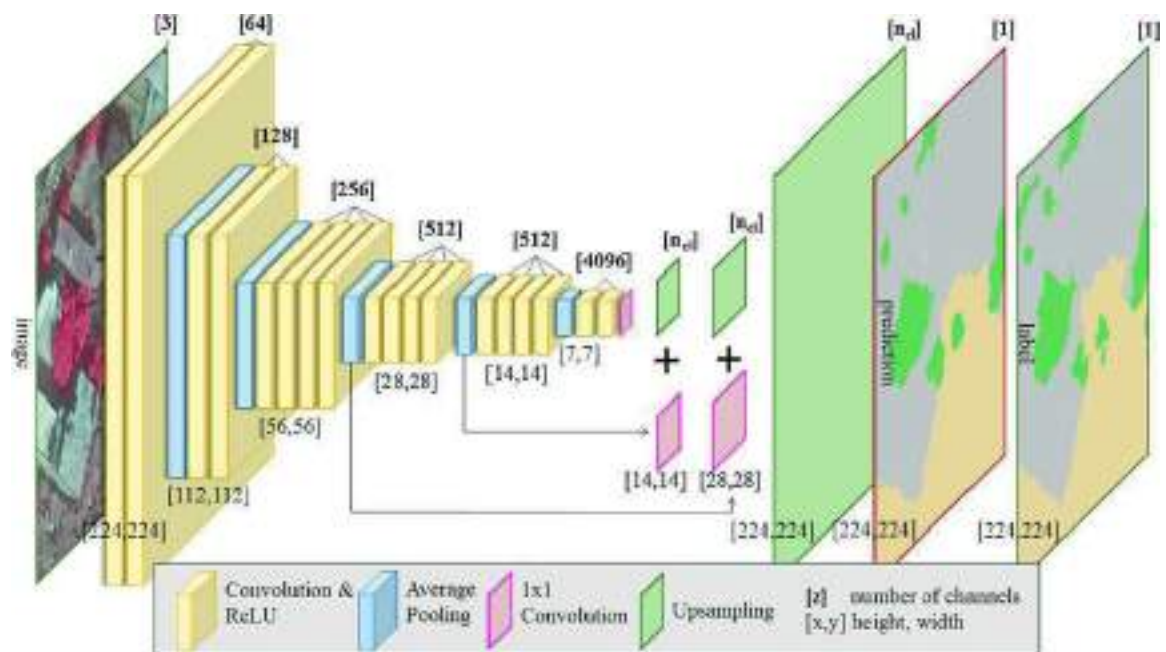
- **Image Classification:** Baseline for many classification tasks.
 - **Object Detection:** Used in R-CNN as a feature extractor.
 - **Semantic Segmentation:** Base model for U-Net, SegNet.
 - **Transfer Learning:** Widely adopted in academia and industry.
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Influence and Subsequent Work

- Inspired standardized kernel sizes (3x3) in many modern CNNs.
- Set foundation for deeper architectures (ResNet, DenseNet).
- Pretrained VGG models remain popular for fine-tuning on custom datasets.



VGG 19



Transfer Learning in Deep Learning

Transfer Learning is a powerful technique where a pre-trained model (usually trained on large datasets like ImageNet) is used as the starting point for a new but related task. The idea is to **leverage previously learned features** to improve learning efficiency and model performance on the new task.

🧠 When to Use Transfer Learning

1. You Have **Less Data**

Case A: Data is **Similar to ImageNet**

- **What to Use:** Pre-trained models like **VGG**, **ResNet**, or **Inception**.
- **How to Use:**
 - Replace the final classification layer with your own task-specific output layer.
 - Use **fine-tuning**: train the new layers (or the full model) using a **low learning rate** so that it adapts to your data without forgetting previous knowledge.

Case B: Data is **Not Similar to ImageNet**

- **What to Use:** Use the pre-trained model as a **feature extractor**.
- **How to Use:**
 - Freeze the convolutional base (do not train).
 - Use the intermediate feature maps as input to your own classifier (e.g., custom dense layers).

- This works because early CNN layers capture general features like edges and textures.
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2. You Have Large Data

Case A: Data is Similar to ImageNet

- **What to Use:** Use a pre-trained model to initialize weights.
- **How to Use:**
 - **Fine-tune the entire model** on your dataset.
 - A moderate learning rate can be used since the model will need to adjust to your specific patterns.

Case B: Data is Not Similar to ImageNet

- **What to Use:** Start with a pre-trained model.
 - **How to Use:**
 - First, use it as a **feature extractor**.
 - Gradually **unfreeze more layers** and fine-tune.
 - With sufficient data, you can **train a new model from scratch** after this warm-up stage.
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Summary

Data Size	Similarity to ImageNet	Strategy
Small	Similar	Fine-tune a pre-trained model
Small	Not Similar	Use pre-trained model as feature extractor
Large	Similar	Fine-tune the entire model
Large	Not Similar	Start with transfer learning, then train custom model

Key Considerations

- **Data Size:**
 - Large datasets allow more training from scratch or deeper fine-tuning.
- **Data Similarity:**
 - If similar to ImageNet (natural images), transfer learning is straightforward.
 - If different (e.g., medical scans, satellite imagery), use the model for feature extraction first.
- **Model Selection:**

- **ResNet** is widely used for general tasks.
 - **DenseNet, Inception**, or other architectures might perform better on specialized tasks.
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Common Steps in a Deep Learning Project

1. **Get the dataset**
2. **Data analysis** and preprocessing
3. **Normalize** the dataset (scaling pixel values, etc.)
4. Use **validation split**
5. Initialize data **generators** (e.g., ImageDataGenerator)
6. Apply **Data Augmentation** (if needed)
7. **Build the model** (transfer learning or custom)
8. **Print model summary**
9. **Compile** the model (choose optimizer, loss, metrics)
10. **Train** the model
11. **Evaluate** the model (on train, validation, test sets)
12. Plot **performance metrics** (accuracy, loss curves, confusion matrix)
13. **Save** the model



Fine-Tuning in Transfer Learning

Fine-tuning refers to taking a **pre-trained model** and further training it on a **smaller, task-specific dataset**. The idea is to adjust the weights slightly to better fit the new data while preserving the general features learned from the large source dataset (e.g., ImageNet).



Steps for Fine-Tuning

1. **Choose a Pre-Trained Model**
Select a model trained on a large dataset like ImageNet. Common choices include VGG, ResNet, Inception, etc.
2. **Freeze Base Layers**
Freeze most of the convolutional base layers to retain previously learned general features like edges and textures.
3. **Replace Output Layers**
Remove the final dense layers and add new layers according to your new task (e.g., change the number of output classes).

4. Compile the Model

Use an appropriate optimizer like `Adam` and a relevant loss function such as `categorical_crossentropy` for multi-class classification.

5. Train Only New Layers

With the base layers frozen, train the new output layers for a few epochs to allow them to initialize well.

6. Unfreeze Some Layers

Gradually unfreeze a few deeper layers in the base model to allow fine-tuning without affecting low-level feature extraction.

7. Fine-Tune with Low Learning Rate

Continue training the model with a low learning rate to gently adjust the weights for the new dataset.

8. Evaluate the Model

Use a validation dataset to ensure the model is improving and not overfitting.

9. Save the Fine-Tuned Model

Store the final model for future inference or deployment.



Optimizing Deep Learning Models

1. Data Optimization

- Data Augmentation
- Normalization and Standardization
- Shuffling the data

2. Efficient Data Pipelines

- Use `prefetching` and parallelism
- Batch your data efficiently

3. Model Architecture Optimization

- Choose a suitable architecture
- Reduce complexity if needed
- Use pre-trained models when possible

4. Hyperparameter Tuning

- Learning rate scheduling
- Optimizer selection
- Regularization techniques (Dropout, L2, etc.)

5. 🖥️ Efficient Computation Techniques

- Mixed precision training
- Gradient clipping
- Gradient accumulation

6. 🖥️ Hardware Utilization

- Use GPUs/TPUs
- Enable multi-GPU and distributed training

7. 📊 Early Stopping & Checkpoints

- Use early stopping to prevent overfitting
- Save model checkpoints regularly

8. 🧠 Advanced Techniques

- Knowledge distillation
- Neural Architecture Search (NAS)
- Ensemble learning

9. 📈 Monitor Training

- Use TensorBoard, MLflow, or Weights & Biases for live tracking

10. ❌ Reduce Overfitting

- Cross-validation
- Data augmentation
- SEED parameter for reproducibility

🔒 Understanding the SEED Parameter

The `seed` parameter ensures **reproducibility** by fixing the random processes in machine learning workflows.

🌟 Purpose

- Ensures same results across runs
- Essential for debugging, testing, and reproducible research

🌟 Where It's Used

- **Data shuffling**
- **Data augmentation**
- **Random weight initialization**
- **Simulation/random sampling**

🌟 How It Works

It initializes the random number generator to a specific state. Given the same seed, the same sequence of "random" values will be generated every time.

🖨️ Example

```
import numpy as np
import tensorflow as tf

np.random.seed(42)
tf.random.set_seed(42)

print(np.random.rand(3)) # Always gives same output
```

⚠️ Impact on Randomness

- Without a seed, outputs differ in every run due to random initialization, shuffling, and augmentation.
- Setting a seed makes the randomness **predictable**, not less random — this ensures consistent results across runs.

📝 Note: Seed Functions in Different Libraries

Each library uses its own method to set the seed:

- `numpy.random.seed()` – NumPy
- `random.seed()` – Python's built-in random module
- `tf.random.set_seed()` – TensorFlow

```
In [1]: !mkdir -p ~/.kaggle
        !cp kaggle.json ~/.kaggle/
```

```
In [2]: !kaggle datasets download -d jtiptj/chest-xray-pneumoniacovid19tuberculosis
```

Warning: Your Kaggle API key is readable by other users on this system! To fix this, you can run 'chmod 600 /root/.kaggle/kaggle.json'

Dataset URL: <https://www.kaggle.com/datasets/jtiptj/chest-xray-pneumoniacovid19tuberculosis>

License(s): other

Downloading chest-xray-pneumoniacovid19tuberculosis.zip to /content

100% 1.73G/1.74G [00:11<00:00, 260MB/s]

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```
In [3]: !unzip /content/chest-xray-pneumoniacovid19tuberculosis.zip -d /content/
```


[illegible]

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[illegible]

[illegible]

[illegible]

```
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```

```
In [4]: import tensorflow as tf
        from tensorflow.keras import layers, models
        from tensorflow.keras.layers import Rescaling

        # For reproducibility
        import numpy as np
        import random
        import os

        seed = 123
        random.seed(seed)
        np.random.seed(seed)
        tf.random.set_seed(seed)
```

```
In [5]: # Directories containing the dataset
        train_dir = '/content/train/' # Update this path to your training data directory
        val_dir = '/content/val/' # Update this path to your validation data directory
        test_dir = '/content/test/' # Update this path to your testing data directory

        # Hyperparameters
        img_width, img_height = 224, 224
        batch_size = 32
        epochs = 10
        num_classes = 4 # Update based on your dataset
```

```
In [6]: # Load the datasets without validation split
        train_dataset = tf.keras.preprocessing.image_dataset_from_directory(
            train_dir,
            image_size=(img_width, img_height),
            batch_size=batch_size
        )
```

Found 6326 files belonging to 4 classes.

```
In [7]: validation_dataset = tf.keras.preprocessing.image_dataset_from_directory(
        val_dir,
        image_size=(img_width, img_height),
```

```
    batch_size=batch_size
)
```

Found 38 files belonging to 4 classes.

```
In [8]: test_dataset = tf.keras.preprocessing.image_dataset_from_directory(
        test_dir,
        image_size=(img_width, img_height),
        batch_size=batch_size
    )
```

Found 771 files belonging to 4 classes.

```
In [9]: # Data Augmentation
data_augmentation = tf.keras.Sequential([
    layers.RandomFlip("horizontal_and_vertical"),
    layers.RandomRotation(0.2),
])
```

```
In [10]: # Apply data augmentation only to the training data
train_dataset = train_dataset.map(lambda x, y: (data_augmentation(x, training=True)
```

```
In [11]: # Prefetch the datasets for better performance
AUTOTUNE = tf.data.AUTOTUNE
train_dataset = train_dataset.prefetch(buffer_size=AUTOTUNE)
validation_dataset = validation_dataset.prefetch(buffer_size=AUTOTUNE)
test_dataset = test_dataset.prefetch(buffer_size=AUTOTUNE)
```

Steps to follow

Get the dataset, Data Analysis

- Normalise the dataset
- Use validation split
- Initialise the generators with proper values.
- Data Augmentation
- Autotune and prefetch.
- Model Building.
- Model summary
- Model compile
- Model training
- Model evaluation on train, test, validation
- Performance plots and metrics.
- Model save

```
In [12]: def build_vgg16(input_shape=(224, 224, 3), num_classes=4):
        model = models.Sequential()

        # Add Input Layer
        model.add(layers.Input(shape=input_shape))
```

```

# Normalization Layer
model.add(Rescaling(1./255))

# Block 1
model.add(layers.Conv2D(64, (3, 3), padding='same', activation='relu'))
model.add(layers.Conv2D(64, (3, 3), padding='same', activation='relu'))
model.add(layers.MaxPooling2D((2, 2), strides=(2, 2)))

# Block 2
model.add(layers.Conv2D(128, (3, 3), padding='same', activation='relu'))
model.add(layers.Conv2D(128, (3, 3), padding='same', activation='relu'))
model.add(layers.MaxPooling2D((2, 2), strides=(2, 2)))

# Block 3
model.add(layers.Conv2D(256, (3, 3), padding='same', activation='relu'))
model.add(layers.Conv2D(256, (3, 3), padding='same', activation='relu'))
model.add(layers.Conv2D(256, (3, 3), padding='same', activation='relu'))
model.add(layers.MaxPooling2D((2, 2), strides=(2, 2)))

# Block 4
model.add(layers.Conv2D(512, (3, 3), padding='same', activation='relu'))
model.add(layers.Conv2D(512, (3, 3), padding='same', activation='relu'))
model.add(layers.Conv2D(512, (3, 3), padding='same', activation='relu'))
model.add(layers.MaxPooling2D((2, 2), strides=(2, 2)))

# Block 5
model.add(layers.Conv2D(512, (3, 3), padding='same', activation='relu'))
model.add(layers.Conv2D(512, (3, 3), padding='same', activation='relu'))
model.add(layers.Conv2D(512, (3, 3), padding='same', activation='relu'))
model.add(layers.MaxPooling2D((2, 2), strides=(2, 2)))

# Fully connected layers
model.add(layers.Flatten())

model.add(layers.Dense(4096, activation='relu'))
model.add(layers.Dropout(0.5))
model.add(layers.Dense(4096, activation='relu'))
model.add(layers.Dropout(0.5))
model.add(layers.Dense(num_classes, activation='softmax')) # Adjust based on n

return model

```

```

In [13]: model = build_vgg16(num_classes=num_classes)
         model.summary()

```

```

Model: "sequential_1"

```

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 224, 224, 3)	0
conv2d (Conv2D)	(None, 224, 224, 64)	1,792
conv2d_1 (Conv2D)	(None, 224, 224, 64)	36,928
max_pooling2d (MaxPooling2D)	(None, 112, 112, 64)	0
conv2d_2 (Conv2D)	(None, 112, 112, 128)	73,856
conv2d_3 (Conv2D)	(None, 112, 112, 128)	147,584
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 128)	0
conv2d_4 (Conv2D)	(None, 56, 56, 256)	295,168
conv2d_5 (Conv2D)	(None, 56, 56, 256)	590,080
conv2d_6 (Conv2D)	(None, 56, 56, 256)	590,080
max_pooling2d_2 (MaxPooling2D)	(None, 28, 28, 256)	0
conv2d_7 (Conv2D)	(None, 28, 28, 512)	1,180,160
conv2d_8 (Conv2D)	(None, 28, 28, 512)	2,359,808
conv2d_9 (Conv2D)	(None, 28, 28, 512)	2,359,808
max_pooling2d_3 (MaxPooling2D)	(None, 14, 14, 512)	0
conv2d_10 (Conv2D)	(None, 14, 14, 512)	2,359,808
conv2d_11 (Conv2D)	(None, 14, 14, 512)	2,359,808
conv2d_12 (Conv2D)	(None, 14, 14, 512)	2,359,808
max_pooling2d_4 (MaxPooling2D)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 4096)	102,764,544
dropout (Dropout)	(None, 4096)	0
dense_1 (Dense)	(None, 4096)	16,781,312
dropout_1 (Dropout)	(None, 4096)	0
dense_2 (Dense)	(None, 4)	16,388

Total params: 134,276,932 (512.23 MB)

Trainable params: 134,276,932 (512.23 MB)

Non-trainable params: 0 (0.00 B)

```
In [14]: model.compile(optimizer='adam',  
                      loss='sparse_categorical_crossentropy', # Use 'categorical_crossentropy'  
                      metrics=['accuracy'])
```

```
In [15]: history = model.fit(  
    train_dataset,  
    validation_data=validation_dataset,  
    epochs=epochs  
)
```

Epoch 1/10

198/198 ————— 208s 854ms/step - accuracy: 0.5583 - loss: 1.3494 - val
_accuracy: 0.2105 - val_loss: 1.8523

Epoch 2/10

198/198 ————— 130s 653ms/step - accuracy: 0.6110 - loss: 1.0605 - val
_accuracy: 0.2105 - val_loss: 1.8139

Epoch 3/10

198/198 ————— 143s 656ms/step - accuracy: 0.6136 - loss: 1.0557 - val
_accuracy: 0.2105 - val_loss: 1.8376

Epoch 4/10

198/198 ————— 129s 651ms/step - accuracy: 0.6099 - loss: 1.0606 - val
_accuracy: 0.2105 - val_loss: 1.8125

Epoch 5/10

198/198 ————— 143s 657ms/step - accuracy: 0.6116 - loss: 1.0583 - val
_accuracy: 0.2105 - val_loss: 1.8058

Epoch 6/10

198/198 ————— 129s 649ms/step - accuracy: 0.6120 - loss: 1.0580 - val
_accuracy: 0.2105 - val_loss: 1.8113

Epoch 7/10

198/198 ————— 130s 653ms/step - accuracy: 0.6102 - loss: 1.0599 - val
_accuracy: 0.2105 - val_loss: 1.7967

Epoch 8/10

198/198 ————— 130s 654ms/step - accuracy: 0.6145 - loss: 1.0545 - val
_accuracy: 0.2105 - val_loss: 1.8227

Epoch 9/10

198/198 ————— 129s 648ms/step - accuracy: 0.6103 - loss: 1.0592 - val
_accuracy: 0.2105 - val_loss: 1.8146

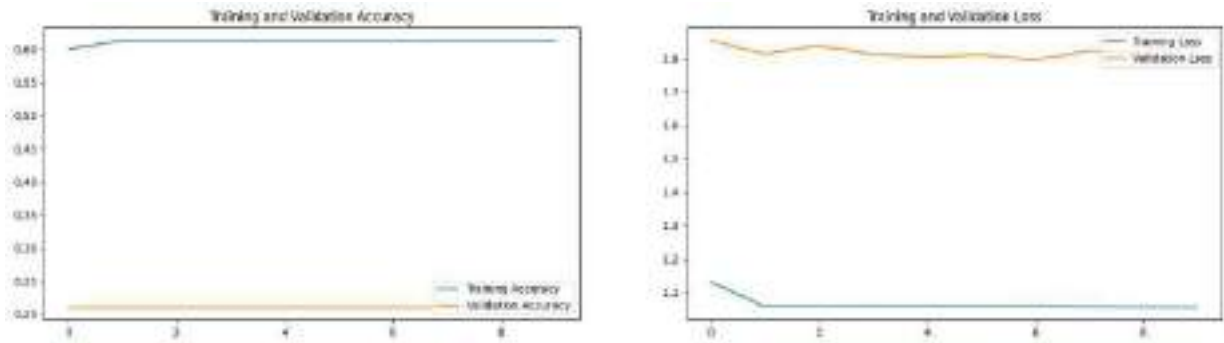
Epoch 10/10

198/198 ————— 129s 650ms/step - accuracy: 0.6112 - loss: 1.0554 - val
_accuracy: 0.2105 - val_loss: 1.8383

```
In [16]: import matplotlib.pyplot as plt  
  
acc = history.history['accuracy']  
val_acc = history.history['val_accuracy']  
loss = history.history['loss']  
val_loss = history.history['val_loss']  
  
epochs_range = range(len(acc))  
  
plt.figure(figsize=(20, 5))  
plt.subplot(1, 2, 1)  
plt.plot(epochs_range, acc, label='Training Accuracy')  
plt.plot(epochs_range, val_acc, label='Validation Accuracy')
```

```
plt.legend(loc='lower right')
plt.title('Training and Validation Accuracy')

plt.subplot(1, 2, 2)
plt.plot(epochs_range, loss, label='Training Loss')
plt.plot(epochs_range, val_loss, label='Validation Loss')
plt.legend(loc='upper right')
plt.title('Training and Validation Loss')
plt.show()
```



```
In [17]: test_loss, test_accuracy = model.evaluate(test_dataset)
print(f'Test Accuracy: {test_accuracy * 100:.2f}%')
```

```
25/25 ————— 9s 353ms/step - accuracy: 0.4901 - loss: 1.2284
Test Accuracy: 50.58%
```

```
In [18]: # Train set predictions
y_train_true = np.concatenate([y for x, y in train_dataset], axis=0)
y_train_pred = model.predict(train_dataset)
y_train_pred = np.argmax(y_train_pred, axis=1)

# Validation set predictions
y_val_true = np.concatenate([y for x, y in validation_dataset], axis=0)
y_val_pred = model.predict(validation_dataset)
y_val_pred = np.argmax(y_val_pred, axis=1)

# Test set predictions
y_test_true = np.concatenate([y for x, y in test_dataset], axis=0)
y_test_pred = model.predict(test_dataset)
y_test_pred = np.argmax(y_test_pred, axis=1)
```

```
198/198 ————— 97s 488ms/step
2/2 ————— 1s 396ms/step
25/25 ————— 8s 297ms/step
```

```
In [19]: from sklearn.metrics import confusion_matrix

# Confusion matrices
cm_train = confusion_matrix(y_train_true, y_train_pred)
cm_val = confusion_matrix(y_val_true, y_val_pred)
cm_test = confusion_matrix(y_test_true, y_test_pred)
```

```
In [20]: import matplotlib.pyplot as plt
import seaborn as sns

# Set up the figure
```

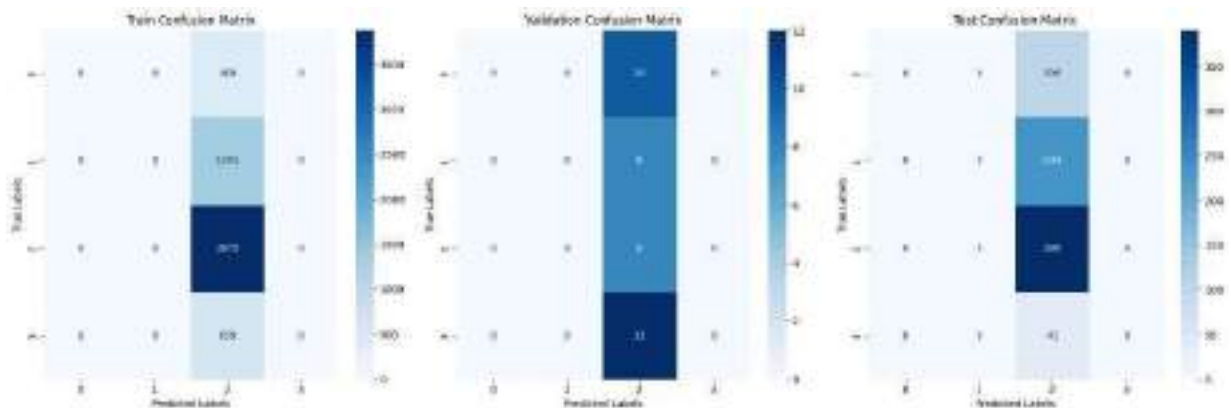
```
fig, axes = plt.subplots(1, 3, figsize=(18, 6))

# Train Confusion Matrix
sns.heatmap(cm_train, annot=True, fmt='d', cmap='Blues', ax=axes[0])
axes[0].set_title('Train Confusion Matrix')
axes[0].set_xlabel('Predicted Labels')
axes[0].set_ylabel('True Labels')

# Validation Confusion Matrix
sns.heatmap(cm_val, annot=True, fmt='d', cmap='Blues', ax=axes[1])
axes[1].set_title('Validation Confusion Matrix')
axes[1].set_xlabel('Predicted Labels')
axes[1].set_ylabel('True Labels')

# Test Confusion Matrix
sns.heatmap(cm_test, annot=True, fmt='d', cmap='Blues', ax=axes[2])
axes[2].set_title('Test Confusion Matrix')
axes[2].set_xlabel('Predicted Labels')
axes[2].set_ylabel('True Labels')

# Adjust Layout
plt.tight_layout()
plt.show()
```



- Very Bad Results

Concept of Transfer Learning

```
In [21]: import tensorflow as tf
from tensorflow.keras.applications import VGG16
from tensorflow.keras import layers, models
from tensorflow.keras.preprocessing import image_dataset_from_directory
from tensorflow.keras.layers import Input
```

```
In [22]: # Load VGG16 with pre-trained ImageNet weights, excluding the top (fully connected)
base_model = VGG16(weights=None,
                    include_top=False,
                    input_tensor=Input(shape=(224, 224, 3)))
```



```
base_model.summary()
```

Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0

Total params: 14,714,688 (56.13 MB)

Trainable params: 14,714,688 (56.13 MB)

Non-trainable params: 0 (0.00 B)

```
In [23]: # Load VGG16 with pre-trained ImageNet weights, excluding the top (fully connected)
base_model = VGG16(weights='imagenet',
                    include_top=False,
                    input_tensor=Input(shape=(img_width, img_height, 3)))
base_model.summary()
```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/vgg16/vgg16_weights_tf_dim_ordering_tf_kernels_notop.h5
 58889256/58889256 ————— 0s 0us/step
 Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer_3 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0

Total params: 14,714,688 (56.13 MB)

Trainable params: 14,714,688 (56.13 MB)

Non-trainable params: 0 (0.00 B)

```

In [24]: # Freeze the layers in the base model
         for layer in base_model.layers:
             layer.trainable = False
  
```

```
In [25]: # Build the model
model = models.Sequential([
    base_model,
    layers.Flatten(),
    layers.Dense(512, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(4, activation='softmax') # Adjust the number of classes
])
```

```
In [26]: model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.0001),
                      loss='sparse_categorical_crossentropy', # or 'categorical_crossentropy'
                      metrics=['accuracy'])
```

Transfer learning with fixed weight in convolutions and training model on FC layers.

```
In [27]: history = model.fit(
    train_dataset,
    validation_data=validation_dataset,
    epochs=10,
)
```

Epoch 1/10

```
/usr/local/lib/python3.11/dist-packages/keras/src/models/functional.py:237: UserWarning: The structure of 'inputs' doesn't match the expected structure.
Expected: ['keras_tensor_348']
Received: inputs=Tensor(shape=(None, 224, 224, 3))
warnings.warn(msg)
```

```

198/198 ————— 92s 448ms/step - accuracy: 0.7397 - loss: 4.9072 - val_
accuracy: 0.7895 - val_loss: 1.1006
Epoch 2/10
198/198 ————— 137s 434ms/step - accuracy: 0.8943 - loss: 0.5848 - val
_accuracy: 0.7632 - val_loss: 0.7795
Epoch 3/10
198/198 ————— 141s 430ms/step - accuracy: 0.9087 - loss: 0.3533 - val
_accuracy: 0.8684 - val_loss: 0.5188
Epoch 4/10
198/198 ————— 142s 429ms/step - accuracy: 0.9107 - loss: 0.2958 - val
_accuracy: 0.7368 - val_loss: 0.6273
Epoch 5/10
198/198 ————— 142s 428ms/step - accuracy: 0.9219 - loss: 0.2489 - val
_accuracy: 0.7895 - val_loss: 0.6762
Epoch 6/10
198/198 ————— 86s 431ms/step - accuracy: 0.9327 - loss: 0.2276 - val_
accuracy: 0.9211 - val_loss: 0.2775
Epoch 7/10
198/198 ————— 141s 424ms/step - accuracy: 0.9248 - loss: 0.2669 - val
_accuracy: 0.8947 - val_loss: 0.2969
Epoch 8/10
198/198 ————— 142s 426ms/step - accuracy: 0.9426 - loss: 0.1745 - val
_accuracy: 0.8947 - val_loss: 0.3254
Epoch 9/10
198/198 ————— 157s 501ms/step - accuracy: 0.9439 - loss: 0.1779 - val
_accuracy: 0.7105 - val_loss: 1.0981
Epoch 10/10
198/198 ————— 95s 477ms/step - accuracy: 0.9378 - loss: 0.2223 - val_
accuracy: 0.7368 - val_loss: 0.5214

```

```

In [28]: import matplotlib.pyplot as plt

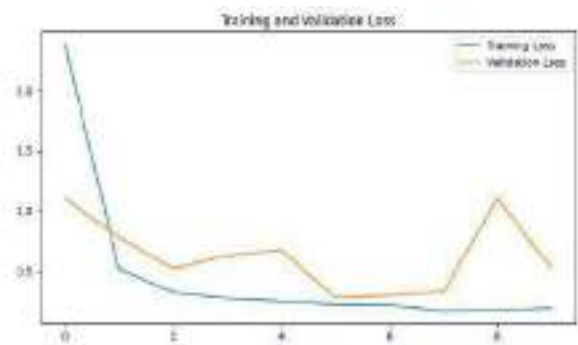
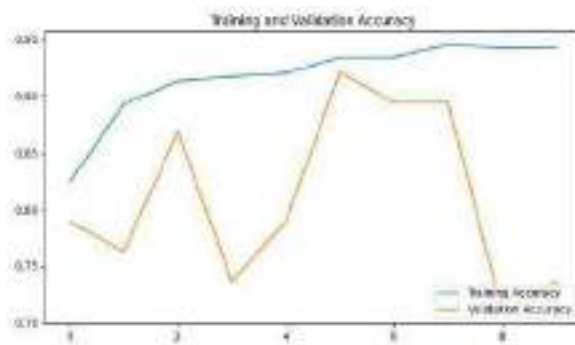
acc = history.history['accuracy']
val_acc = history.history['val_accuracy']
loss = history.history['loss']
val_loss = history.history['val_loss']

epochs_range = range(len(acc))

plt.figure(figsize=(20, 5))
plt.subplot(1, 2, 1)
plt.plot(epochs_range, acc, label='Training Accuracy')
plt.plot(epochs_range, val_acc, label='Validation Accuracy')
plt.legend(loc='lower right')
plt.title('Training and Validation Accuracy')

plt.subplot(1, 2, 2)
plt.plot(epochs_range, loss, label='Training Loss')
plt.plot(epochs_range, val_loss, label='Validation Loss')
plt.legend(loc='upper right')
plt.title('Training and Validation Loss')
plt.show()

```



```
In [29]: test_loss, test_accuracy = model.evaluate(test_dataset)
print(f'Test Accuracy: {test_accuracy * 100:.2f}%')
```

25/25 ————— 7s 276ms/step - accuracy: 0.7995 - loss: 0.7719
Test Accuracy: 79.77%

```
In [30]: # Train set predictions
y_train_true = np.concatenate([y for x, y in train_dataset], axis=0)
y_train_pred = model.predict(train_dataset)
y_train_pred = np.argmax(y_train_pred, axis=1)

# Validation set predictions
y_val_true = np.concatenate([y for x, y in validation_dataset], axis=0)
y_val_pred = model.predict(validation_dataset)
y_val_pred = np.argmax(y_val_pred, axis=1)

# Test set predictions
y_test_true = np.concatenate([y for x, y in test_dataset], axis=0)
y_test_pred = model.predict(test_dataset)
y_test_pred = np.argmax(y_test_pred, axis=1)
```

```
/usr/local/lib/python3.11/dist-packages/keras/src/models/functional.py:237: UserWarning: The structure of 'inputs' doesn't match the expected structure.
Expected: ['keras_tensor_348']
Received: inputs=Tensor(shape=(32, 224, 224, 3))
warnings.warn(msg)
```

198/198 ————— 91s 453ms/step
2/2 ————— 1s 403ms/step
25/25 ————— 7s 261ms/step

```
In [38]: from sklearn.metrics import confusion_matrix
```

```
# Confusion matrices
cm_train = confusion_matrix(y_train_true, y_train_pred)
cm_val = confusion_matrix(y_val_true, y_val_pred)
cm_test = confusion_matrix(y_test_true, y_test_pred)
```

```
In [39]: import matplotlib.pyplot as plt
import seaborn as sns

# Set up the figure
fig, axes = plt.subplots(1, 3, figsize=(18, 6))

# Train Confusion Matrix
sns.heatmap(cm_train, annot=True, fmt='d', cmap='Blues', ax=axes[0])
```

```

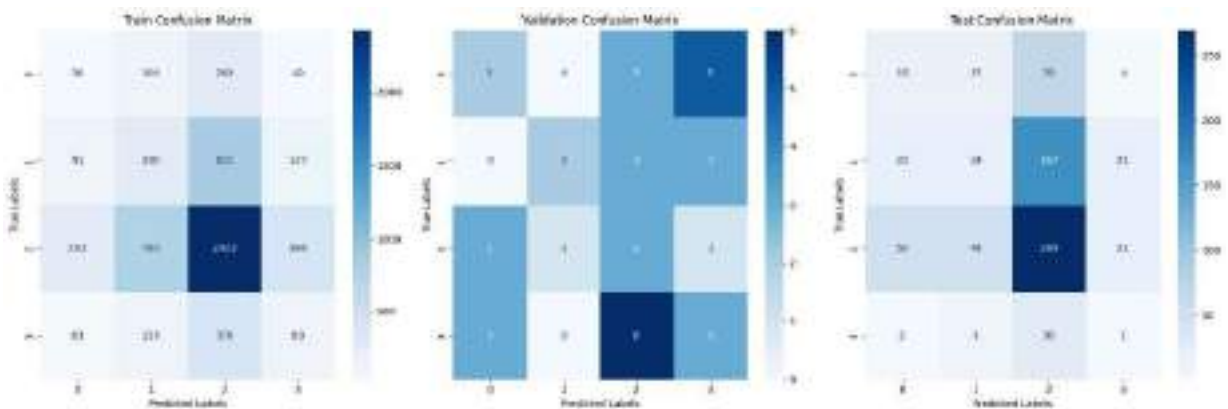
axes[0].set_title('Train Confusion Matrix')
axes[0].set_xlabel('Predicted Labels')
axes[0].set_ylabel('True Labels')

# Validation Confusion Matrix
sns.heatmap(cm_val, annot=True, fmt='d', cmap='Blues', ax=axes[1])
axes[1].set_title('Validation Confusion Matrix')
axes[1].set_xlabel('Predicted Labels')
axes[1].set_ylabel('True Labels')

# Test Confusion Matrix
sns.heatmap(cm_test, annot=True, fmt='d', cmap='Blues', ax=axes[2])
axes[2].set_title('Test Confusion Matrix')
axes[2].set_xlabel('Predicted Labels')
axes[2].set_ylabel('True Labels')

# Adjust Layout
plt.tight_layout()
plt.show()

```



Even more finetuning

```

In [40]: # Load VGG16 with pre-trained ImageNet weights, excluding the top (fully connected)
base_model = VGG16(weights='imagenet',
                    include_top=False,
                    input_tensor=Input(shape=(224, 224, 3)))

# Freeze the first few layers
for layer in base_model.layers[:13]: # Freeze first 15 layers, adjust as needed
    layer.trainable = False

# Unfreeze the remaining layers
for layer in base_model.layers[13:]:
    layer.trainable = True

In [34]: base_model.summary()

```

Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer_5 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0

Total params: 14,714,688 (56.13 MB)

Trainable params: 9,439,232 (36.01 MB)

Non-trainable params: 5,275,456 (20.12 MB)

```
In [35]: # Load VGG16 with pre-trained ImageNet weights, excluding the top (fully connected)
base_model = VGG16(weights='imagenet',
                    include_top=False,
                    input_tensor=Input(shape=(224, 224, 3)))

# Load VGG16 with pre-trained ImageNet weights, excluding the top (fully connected)
base_model2 = VGG16(weights='imagenet',
                    include_top=True,
                    input_tensor=Input(shape=(224, 224, 3)))
```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/vgg16/vgg16_weights_tf_dim_ordering_tf_kernels.h5
553467096/553467096 ————— 6s 0us/step

```
In [36]: base_model.summary()
```

Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer_6 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0

Total params: 14,714,688 (56.13 MB)

Trainable params: 14,714,688 (56.13 MB)

Non-trainable params: 0 (0.00 B)

```
In [37]: base_model2.summary()
```


Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer_7 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
fc1 (Dense)	(None, 4096)	102,764,544
fc2 (Dense)	(None, 4096)	16,781,312
predictions (Dense)	(None, 1000)	4,097,000

Total params: 138,357,544 (527.79 MB)

Trainable params: 138,357,544 (527.79 MB)

Non-trainable params: 0 (0.00 B)

